



# RIDDHIMAN MOULICK

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## EDUCATION

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|----------------------------------------------------------------------------|----------------|
| <b>Indian Institute of Technology, Kharagpur</b>   <i>CGPA: 9.04/10.00</i> | 2020 – Present |
| Major: Electrical Engineering                                              |                |
| Minor: Computer Science & Engineering                                      |                |
| Micro: Artificial Intelligence and Applications                            |                |
| <b>City International School, Pune</b>   <i>Grade: 96.8%</i>               | 2018 – 2020    |
| Higher Secondary (CBSE)                                                    |                |
| <b>The Bishop's School, Undri, Pune</b>   <i>Grade: 98.4%</i>              | 2006 – 2018    |
| Secondary School (ICSE)                                                    |                |

## WORK EXPERIENCE

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| <b>Convolutional Prompting for Continual Learning</b>   IIT Kharagpur                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Aug' 2023 – Dec' 2023 |
| Bachelor Thesis Project   <i>Supervisor: Prof. Abir Das (CSE, IIT Kharagpur)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                       |
| <ul style="list-style-type: none"><li>Devised a novel convolution-based, <i>on-the-fly prompting</i> technique to overcome the challenge of <i>catastrophic forgetting</i> in the class-incremental continual learning setup in vision models.</li><li>Experimented with various task-similarity metrics based on <i>class-label</i>, <i>image</i> and <i>class-attribute embeddings</i> and chose the class attributes' similarity metric for a parameter-efficient prompt pool expansion.</li><li>Achieved a performance improvement of <math>\sim 3\%</math> over SOTA approaches on benchmark datasets while also <i>reducing the parameter overhead</i> by <math>\sim 50\%</math> (Paper accepted at CVPR 2024 Conference).</li></ul> |                       |
| <b>Becky Recommendations</b>   Oracle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | May 2023 – Jul' 2023  |
| Cloud Applications Developer Intern                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                       |
| <ul style="list-style-type: none"><li>Added the user action recommendation feature in <i>Becky</i>, a digital assistant, to automatically initiate a contextual conversation with users, and suggest actions based on their possible requirements.</li><li>Fetched various users' pods and analyzed the data obtained from these, to understand the prospective needs of the user and to alert the user about delayed pods or action updates via mail.</li></ul>                                                                                                                                                                                                                                                                           |                       |

## PROJECTS AND RESEARCH

|                                                                                                                                                                                                                                                           |           |
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| <b>Computing Game-Theoretic Equilibria</b>   <i>Course Project: Algorithmic Game Theory</i>                                                                                                                                                               | Fall 2022 |
| <ul style="list-style-type: none"><li>Acquired all the parameterized inputs in normal form games by parsing the input from Gambit GUI</li><li>Computed and enlisted the various strategies for Dominant strategy equilibria and Nash equilibria</li></ul> |           |
| <b>Navigating a Frozen Lake with Stochastic Wind</b>   <i>Course Project: Reinforcement Learning</i>                                                                                                                                                      | Fall 2023 |
| <ul style="list-style-type: none"><li>Applied Policy Evaluation, Iteration, and Monte-Carlo Algorithms for agent navigation</li><li>Visualized the learning iterations and the frozen lake environment using the OpenAI <i>gym</i> package</li></ul>      |           |

## COURSEWORK

Computer Architecture and Organisation, Algorithms-I, Deep Learning Foundations and Applications, Reinforcement Learning, Algorithmic Game Theory, AI for Economics, Information Security, Database Management Systems

## SKILLS

- Programming Languages:** Python, C/C++, Java, SQL, Verilog, Assembly,  $\text{\LaTeX}$
- Tools & Libraries:** PyTorch, Tensorflow, Keras, OpenCV, NumPy, Matplotlib, Pandas, Matlab, Simulink, Stateflow

## AWARDS AND ACHIEVEMENTS

- Achieved an **All India Rank 6** and city rank 1 in ICSE Class X with an aggregate score of **98.4%**
- Awarded the **Silver Certificate** for securing **AIR 233** in Technothon-2016, conducted by IIT Guwahati
- Secured a Department change by being among the **top 5%** of the batch out of 1800+ first-year students

[\*CV last updated on February 29, 2024]